

COMPUTER ENGINEERING / PORTFOLIO REPORT

Build with intent.

Practical software where automation, interfaces,
and safety have to work together.

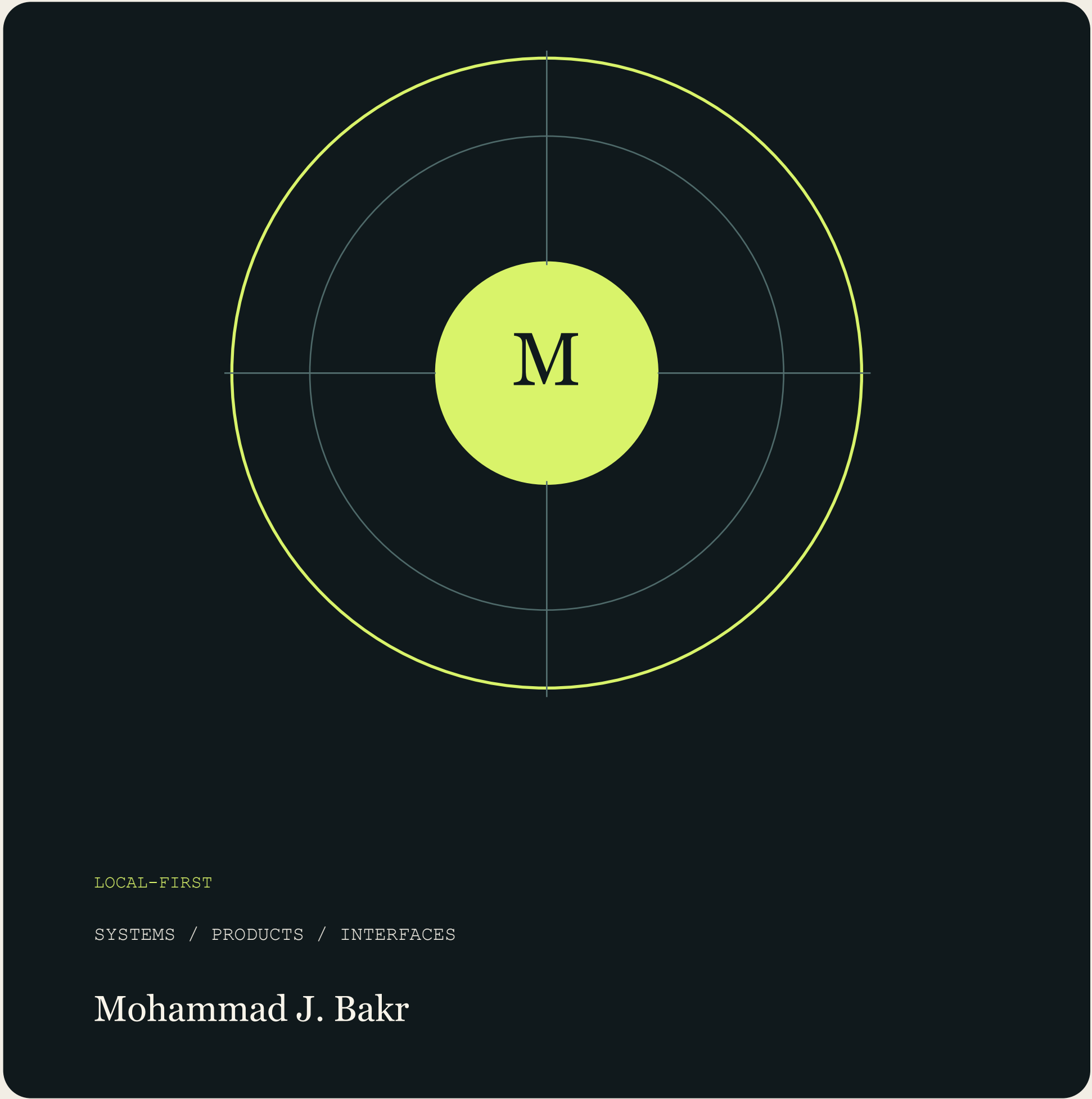
PYTHON

TYPESCRIPT

REACT

SQLITE

EMBEDDED



THE THROUGHLINE

One engineering instinct. Many surfaces.

The work moves from embedded safety hardware to local AI systems, Arabic-first products, automation bots, and motion compositions.

01

SYSTEMS

State, policy, routing, and auditability.

- AgenticOS ☐
- Operations Explorer ☐
- Alfred Clean ☐

02

PRODUCTS

Real workflows shaped around real users.

- Rizq POS ☐
- Job Search Dashboard ☐
- Arabic News Bot ☐

03

SURFACES

Interfaces and media that make ideas legible.

- Malham Line ☐
- Remotion compositions ☐
- Little Lanterns ☐

BUILD → EXPLAIN → VERIFY → SHIP SAFELY

01 / FEATURED CASE

AgenticOS

A personal agent operating system with a safety boundary.

THE IDEA

Give agents useful capabilities without giving them silent authority over the outside world.

THE PROOF

16

unit tests pass

SHA-256

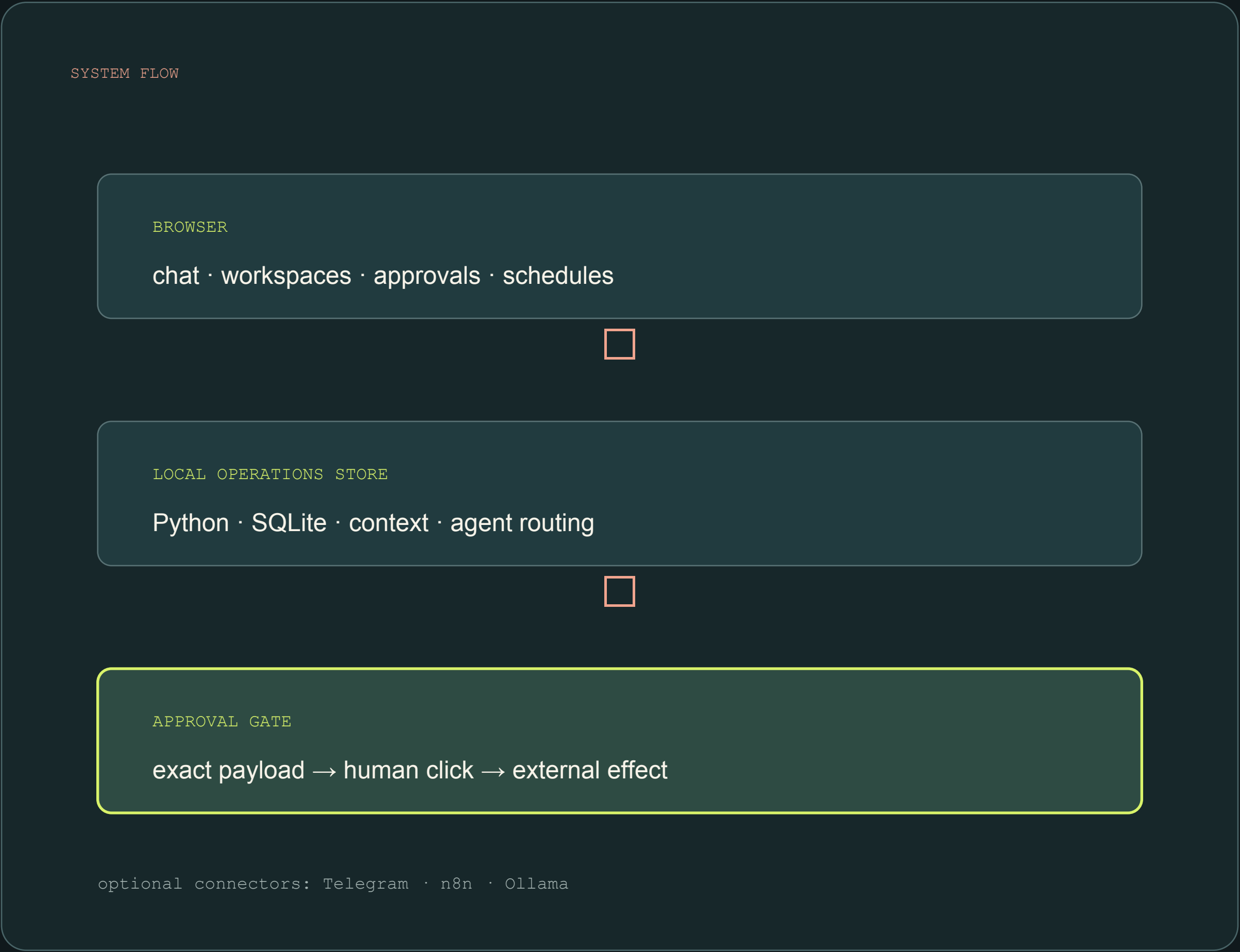
audit chain

127.0.0.1

loopback default

PUBLIC SNAPSHOT

runtime data excluded

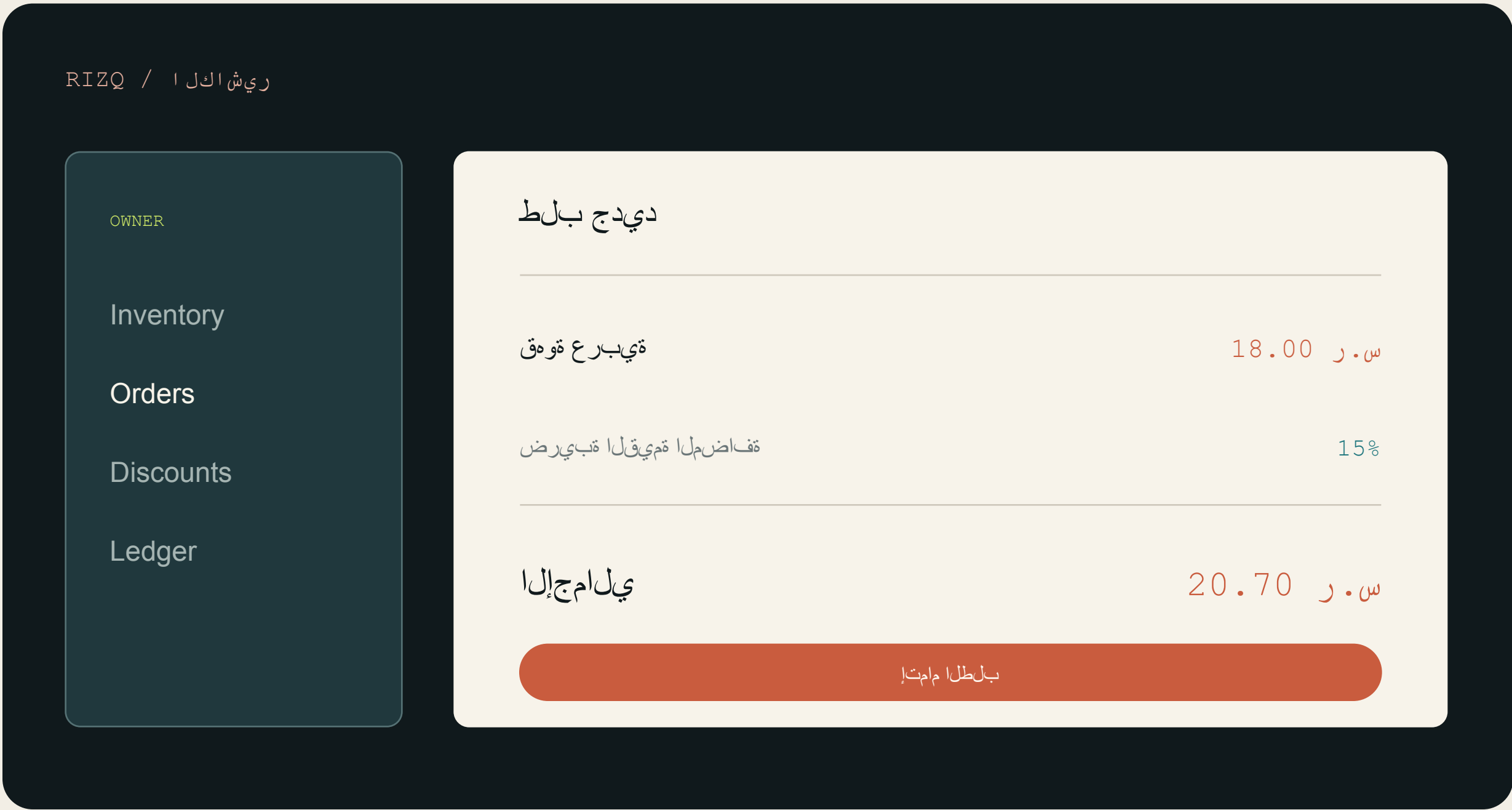


github.com/M7mdbkr/agenticos

02 / PRODUCT CASE

Rizq POS

Arabic-first RTL point of sale for cafés and restaurants.



THE PRODUCT STORY

A real operational surface,
not just a dashboard mockup.

- Cashier + owner flows.
- Browser-local SQLite/WASM.
- Inventory, receipts, CSV export.
- Arabic typography and RTL layout.

READINESS NOTE

Build passes. Harden before real customer data.

REACT

TYPESCRIPT

VITE

SQL.JS

github.com/M7mdbkr/rizq-pos-demo

From signal to useful output.

The interesting part is the chain: fetch, normalize, store, translate, explain, and send only what the user can inspect.

ARABIC NEWS BOT

RSS → Arabic → Telegram

- multi-feed polling
- SQLite deduplication
- owner-only commands
- weekly report + video script

CRYPTO SIGNAL ANALYZER

Data, with the caveat attached.

- CoinGecko market data
- volatility + drawdown + rank
- 7d / 30d momentum
- informational, not a promise

Smart Black Box

Graduation project · ESP32 vehicle safety prototype.

THE HARDWARE LOOP

01

SENSE

MPU6050 · MQ-135 · GPS · OBD-II

02

DECIDE

ESP32 · configurable thresholds

03

RESPOND

SD log · SMS · Google Maps link

WHY IT MATTERS

A low-cost, locally engineered record of impact, location, and cabin conditions.

PROJECT LEAD

Mohammad J. Bakr

14-member team · Taif University

C++

ESP32

OBD-II

GPS / GSM

GITHUB STATUS

private by choice · graduation team project · access on request

Make the system legible.

A selection of surfaces built around hierarchy, timing, and context.

01	MALHAM LINE	ROUTE PLANNING	bilingual stops · timing · SVG route	
02	JOB SEARCH	WORKFLOW UX	matches · drafts · replies · applications	
03	TECH NEWS	MOTION SYSTEM	Arabic captions · voice layers · 9:16	
04	LITTLE LANTERNS	PRODUCTION PIPELINE	storyboard · hooks · QA manifest	

PUBLIC SNAPSHOT RULE

code is shareable; imported media, generated audio, and private footage stay out until rights are clear.

Let’s build the next useful system.

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Saudi Arabia · Arabic native · English fluent

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github.com/M7mdbkr

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OPEN TO ENGINEERING ROLES

Thank you for looking.

github.com/M7mdbkr/mohammad-portfolio